

MAIN_CRAFT TASK 2

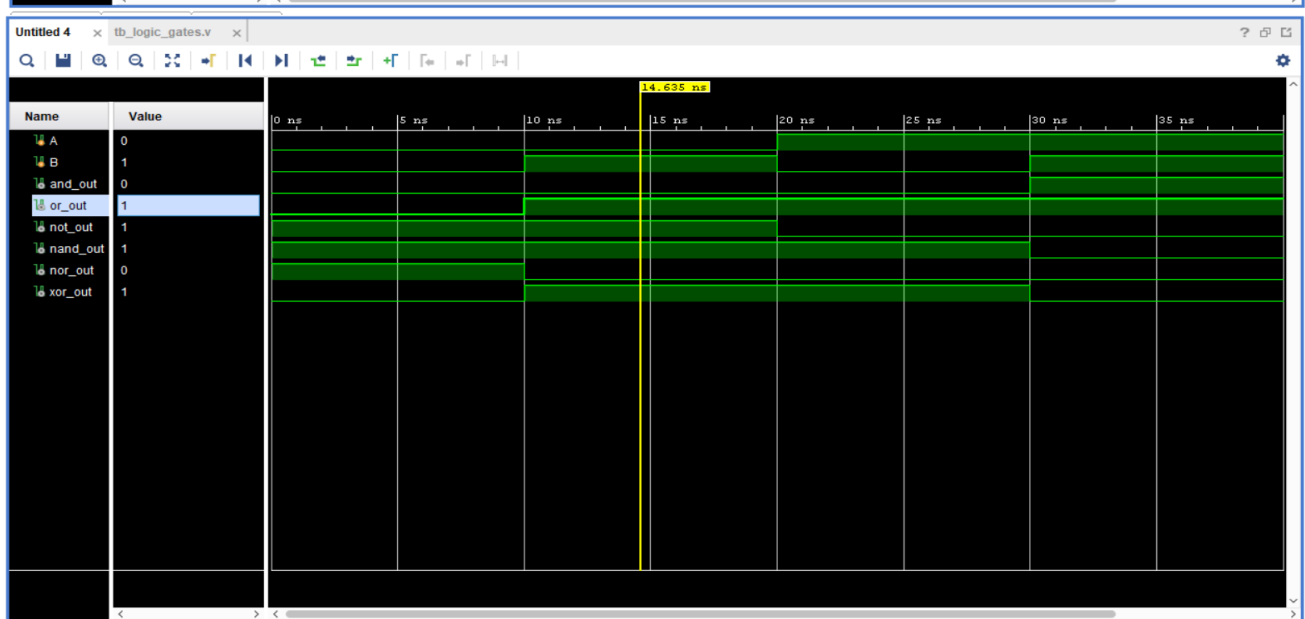
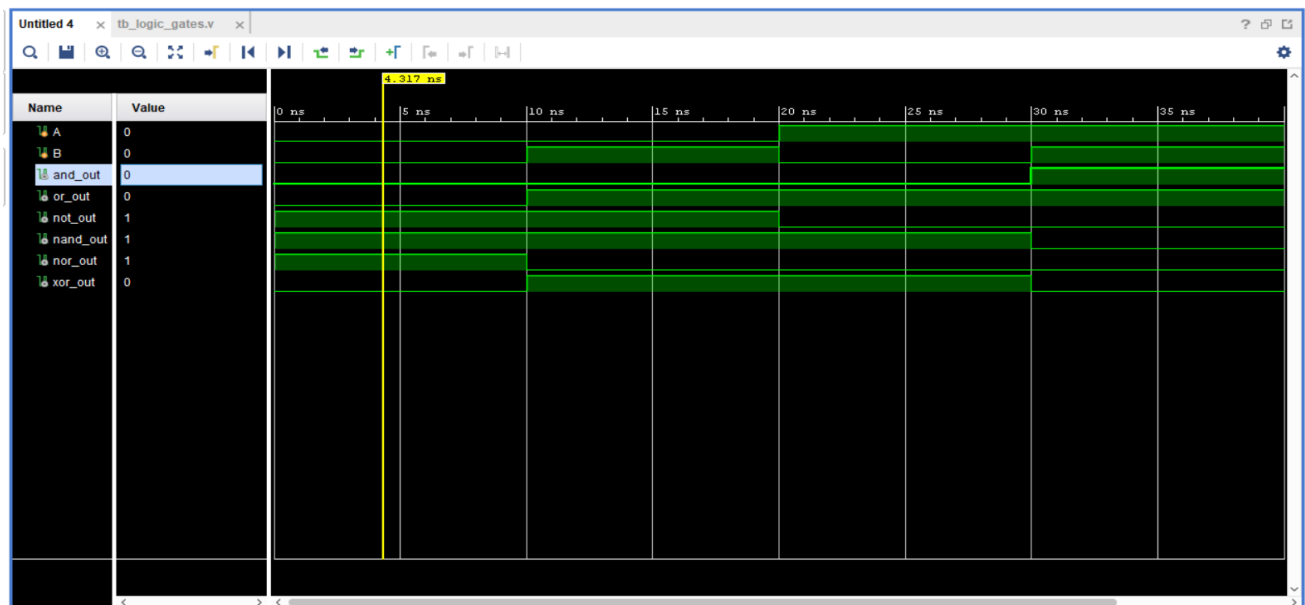
Verilog HDL and RTL Design of Basic Combinational Circuits

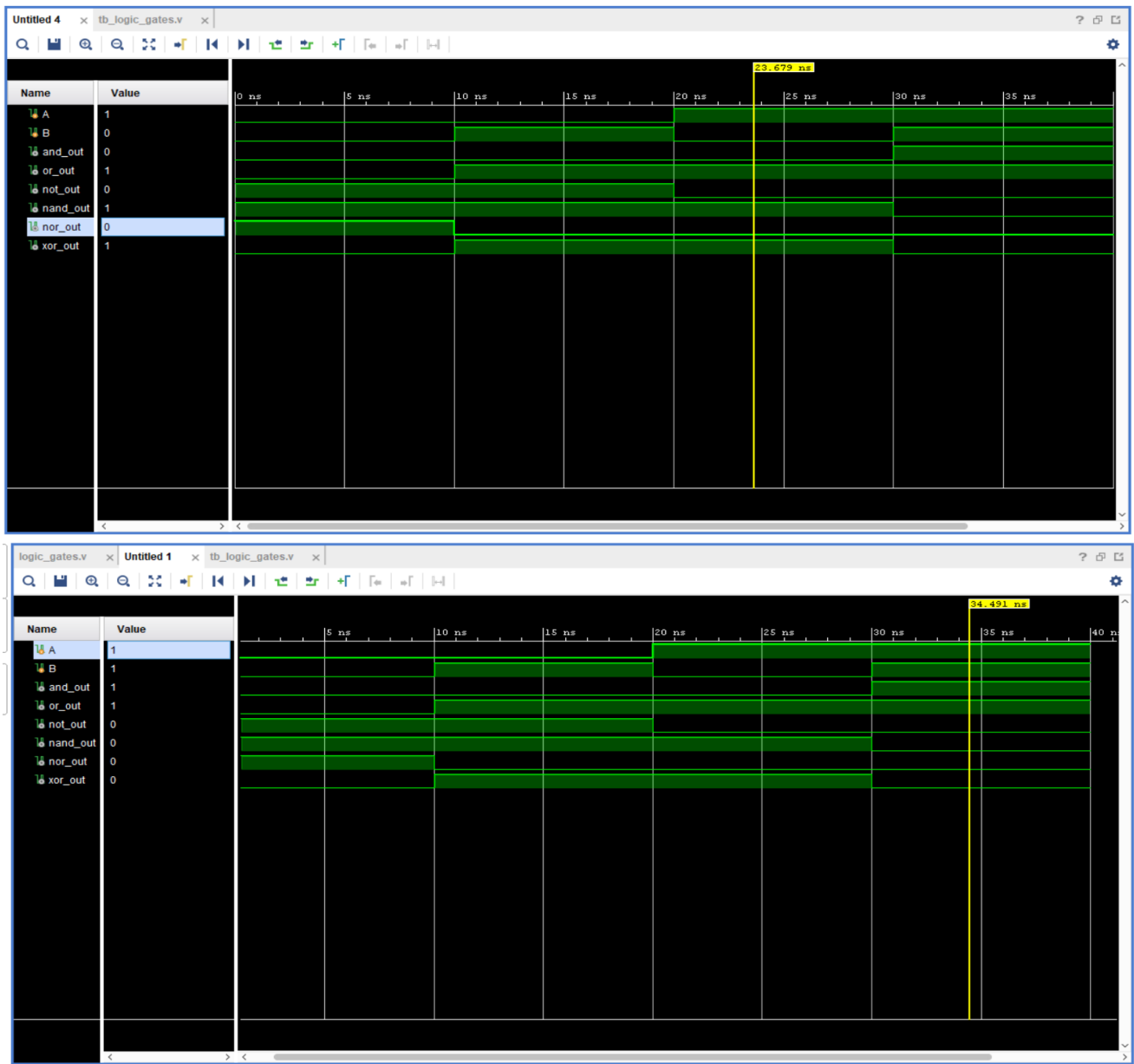
This document explains the design logic and simulation results of the Verilog modules implemented during Task 2. The implemented modules include Logic Gates, Half Adder and Full Adder. Each design was verified using an independent testbench and waveform simulation in Vivado.

1. Logic Gates

Basic combinational logic gates were implemented using continuous assignment statements in Verilog. The functionality was verified using all input combinations.

A	B	AND	OR	NOT(A)	NAND	NOR	XOR
0	0	0	0	1	1	1	0
0	1	0	1	1	1	0	1
1	0	0	1	0	1	0	1
1	1	1	1	0	0	0	0

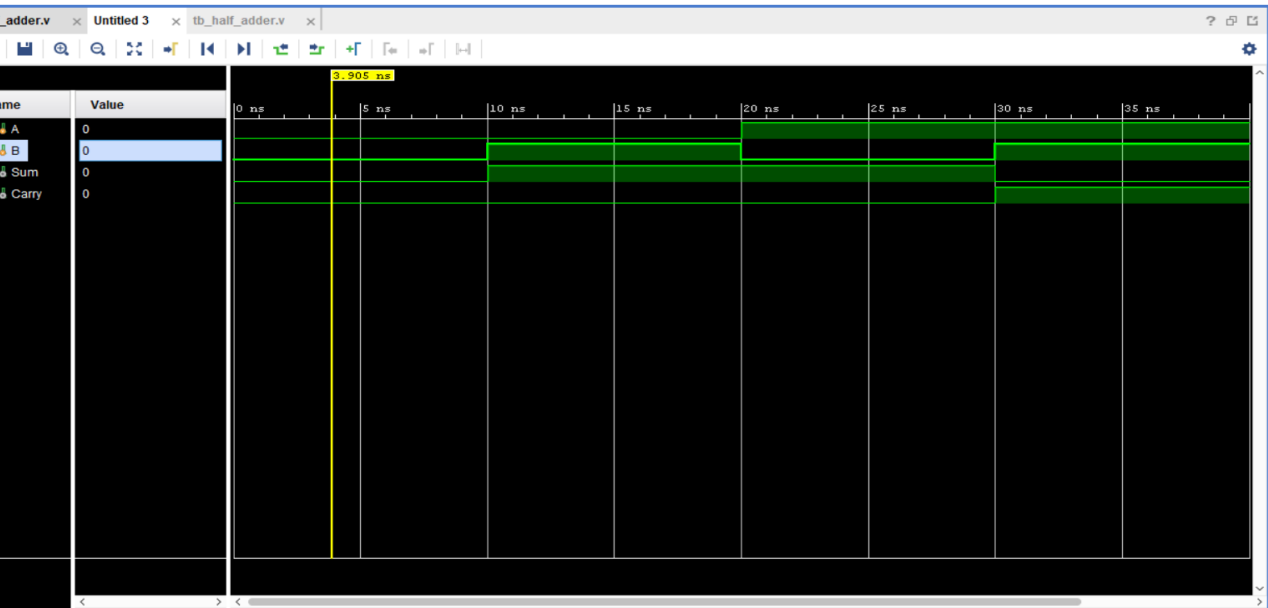


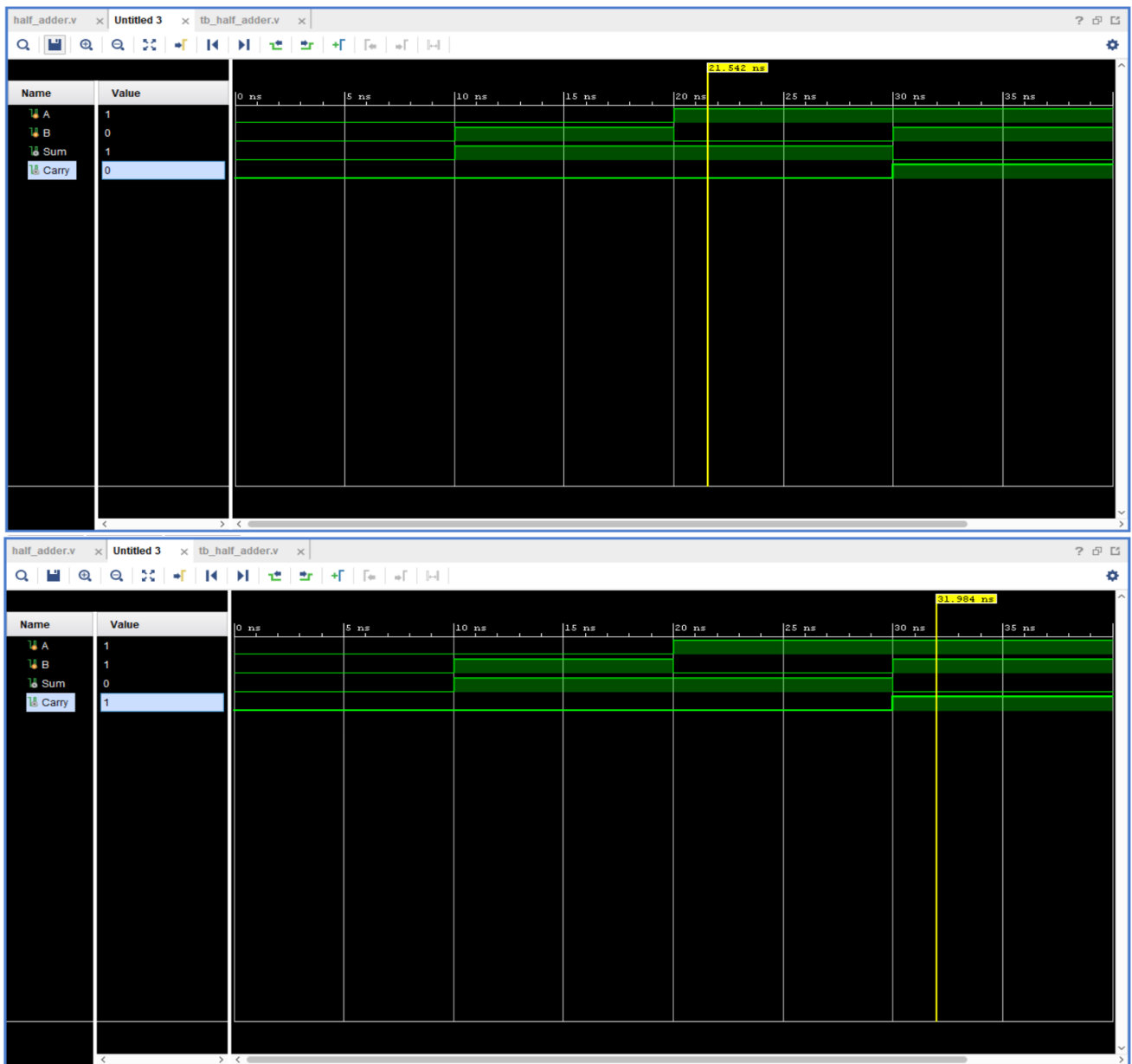


2. Half Adder

Half Adder performs addition of two single-bit binary inputs. Sum is generated using XOR operation while Carry is generated using AND operation.

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

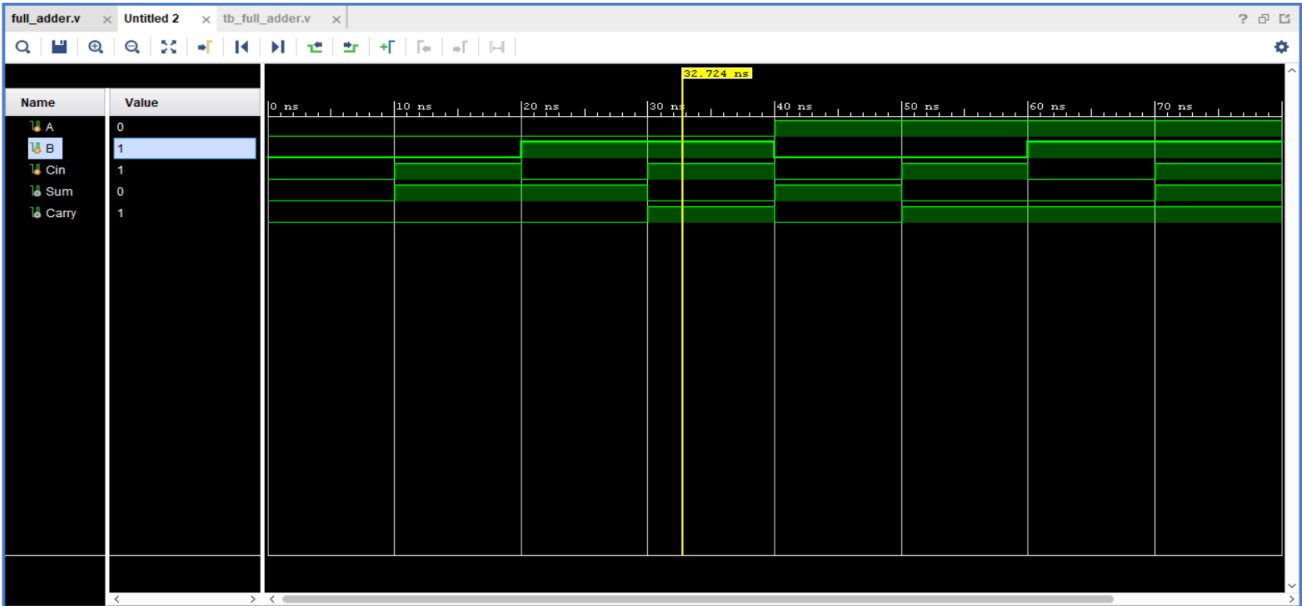
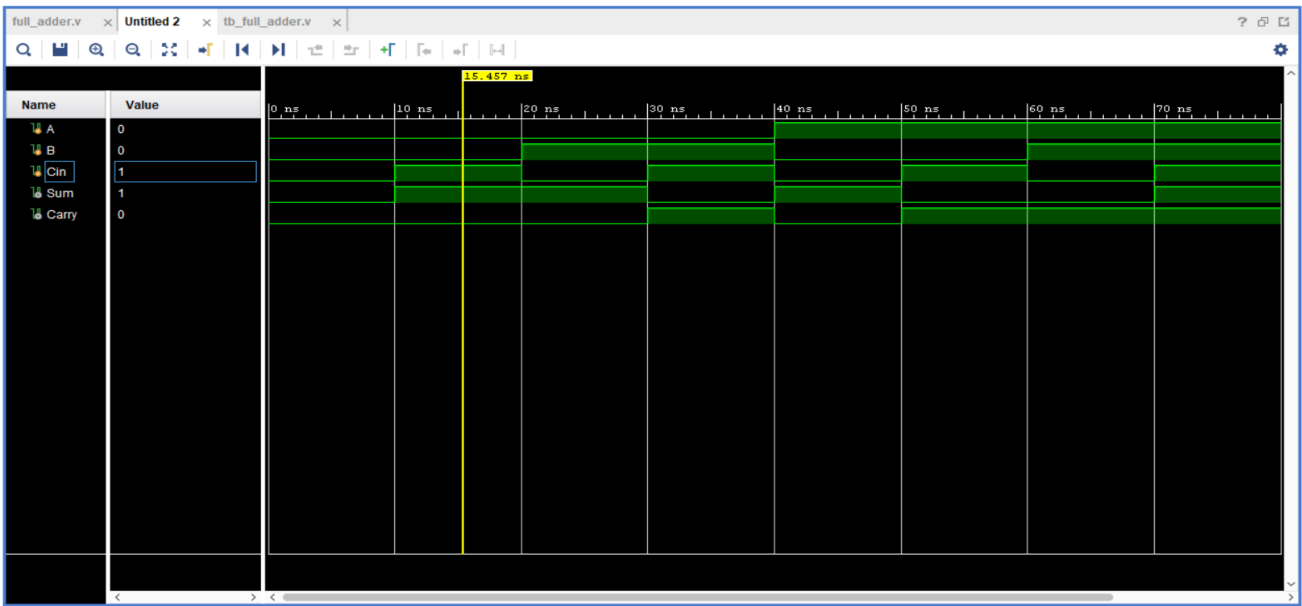
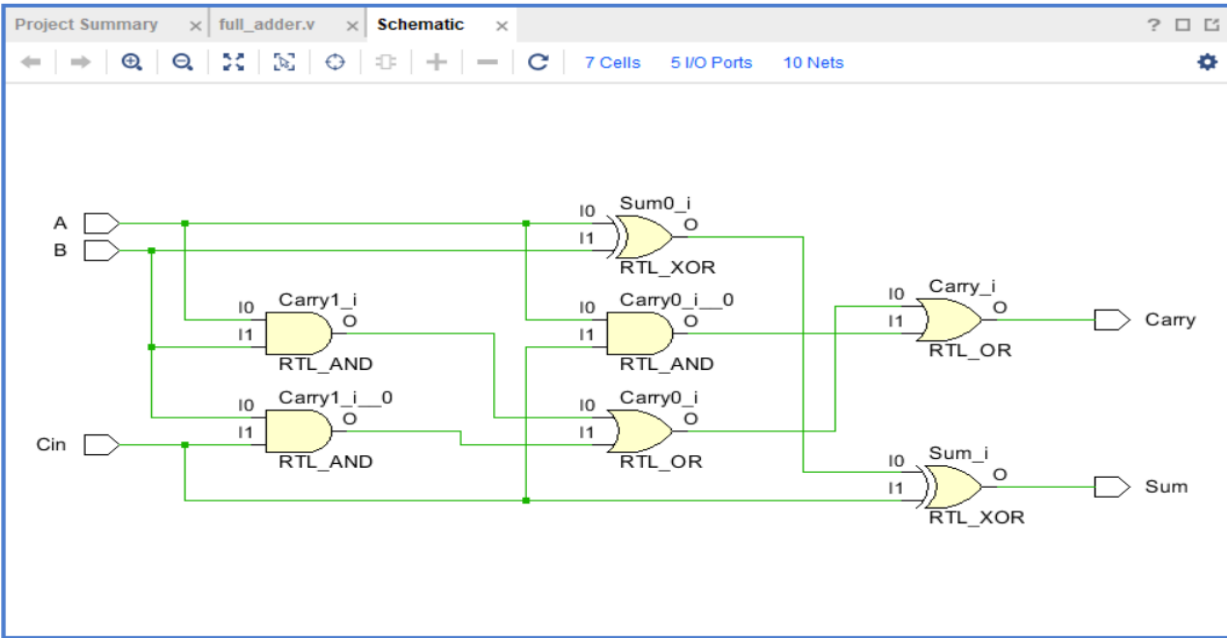


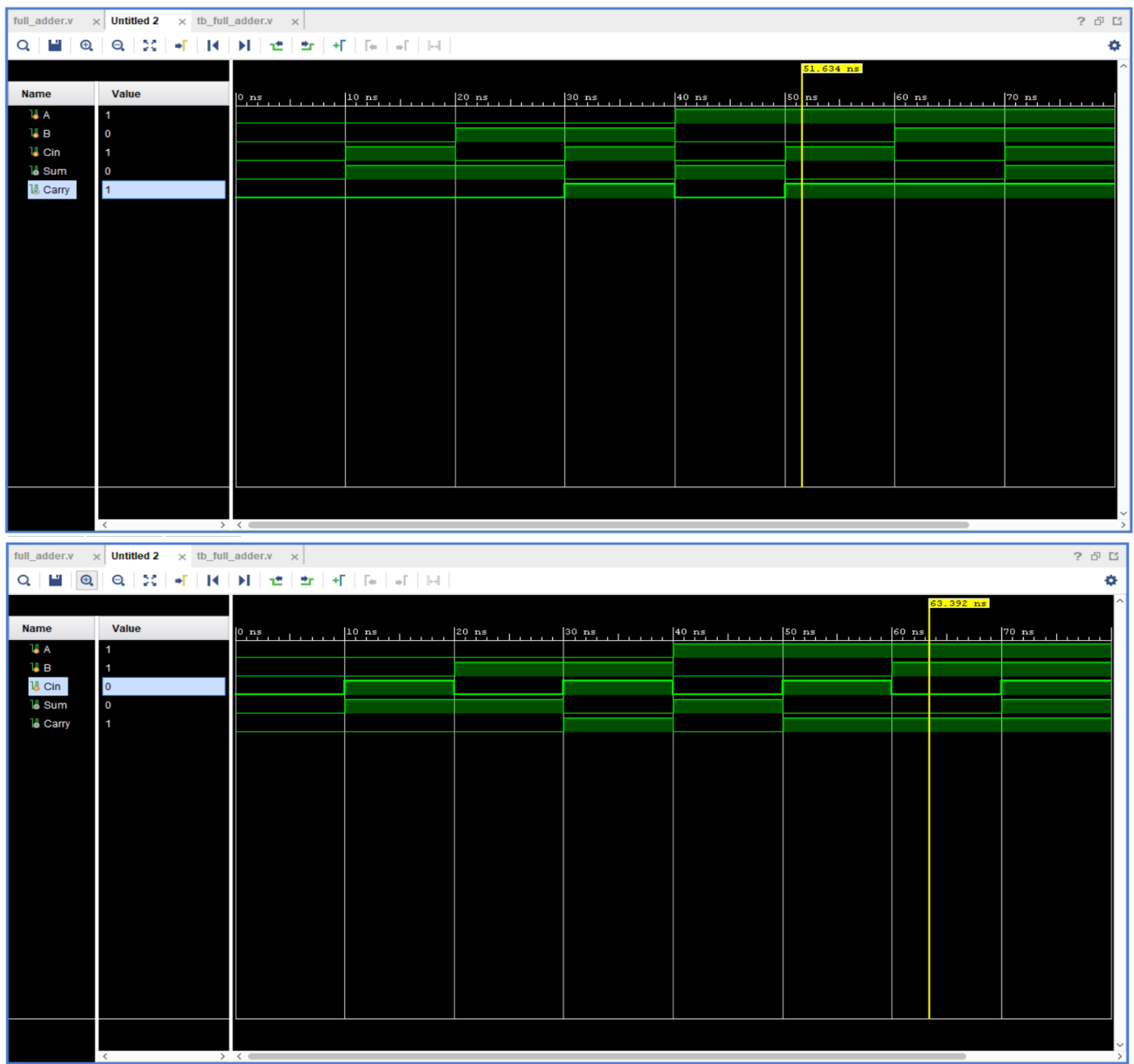


3. Full Adder

Full Adder performs addition of three one-bit inputs (A, B and Carry-in). The Sum output is obtained using XOR logic while Carry-out is produced using OR and AND combinations.

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1





Conclusion

All Verilog modules were successfully designed, simulated and verified. The generated waveforms matched the expected truth tables, confirming the correctness of the RTL implementations.